

Factorisation Method for the Sech-Squared Potential

Cambridge Quantum Mechanics, Example Sheet 1, Question 8; accompanies David Tong's notes.

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Quantum Mechanics

Sketch the potential

$$U(x) = -\frac{\hbar^2}{m} \operatorname{sech}^2 x. \quad (1)$$

Show that the time-independent Schrödinger equation for a particle in this potential can be written

$$\hat{A}^\dagger \hat{A} \chi = (\mathcal{E} + 1) \chi, \quad (2)$$

where $\mathcal{E} = 2mE/\hbar^2$ and

$$\hat{A} = \frac{d}{dx} + \tanh x, \quad \hat{A}^\dagger = -\frac{d}{dx} + \tanh x. \quad (3)$$

Show, by integrating by parts, that for any normalised wavefunction χ ,

$$\int_{-\infty}^{\infty} \chi^* \hat{A}^\dagger \hat{A} \chi dx = \int_{-\infty}^{\infty} (\hat{A} \chi)^* (\hat{A} \chi) dx, \quad (4)$$

and deduce that the eigenvalues of $\hat{A}^\dagger \hat{A}$ are non-negative. Hence show that the ground state (with lowest energy) has $\mathcal{E} \geq -1$. Show that a wavefunction $\chi_0(x)$ is an energy eigenstate with $\mathcal{E} = -1$ iff

$$\frac{d\chi_0}{dx} + \tanh x \chi_0 = 0. \quad (5)$$

Find and sketch $\chi_0(x)$.

Solution

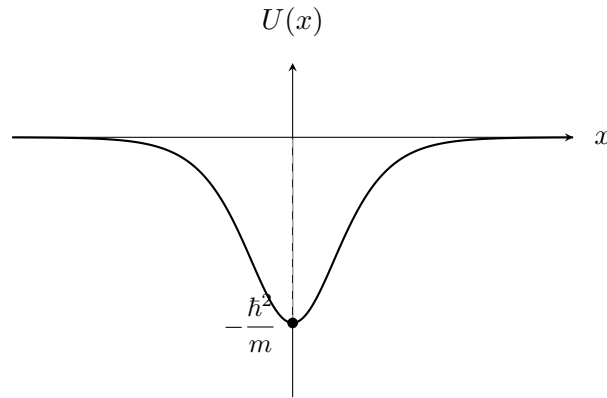


Figure 1. The sech-squared potential (1), a reflectionless well of depth $\hbar^2/m$ centred at $x = 0$, decaying rapidly as $x \rightarrow \pm\infty$.

The Hamiltonian can be written as

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - \frac{\hbar^2}{m} \operatorname{sech}^2 x \quad (6)$$

$$= \frac{\hbar^2}{2m} \left(-\frac{d^2}{dx^2} - 2 + \tanh^2 x \right) \quad (7)$$

$$= \frac{\hbar^2}{2m} \left(\frac{d}{dx} + \tanh x \right) \left(-\frac{d}{dx} + \tanh x \right) - \frac{\hbar^2}{m}. \quad (8)$$

Thus, we have

$$\hat{A}^\dagger \hat{A} \chi = (\mathcal{E} + 1) \chi. \quad (9)$$

Rather trivially following from integrating by part, we find

$$\int dx \chi^* \hat{A}^\dagger \hat{A} \chi = \int dx (\hat{A} \chi)^* (\hat{A} \chi), \quad (10)$$

justifying that $\hat{A}^\dagger$ is indeed the Hermitian conjugate to $\hat{A}$. Given an eigenstate of $\hat{A}^\dagger \hat{A}$, $|a\rangle$, with eigenvalue a . We have

$$\langle a | \hat{A}^\dagger \hat{A} | a \rangle = a \langle a | a \rangle = \| \hat{A} | a \rangle \|^2 \geq 0 \implies a \geq 0, \quad (11)$$

which implies that $\mathcal{E} + 1 \geq 0$. For the ground state with energy $\mathcal{E} = -1$, we have

$$\hat{A}^\dagger \hat{A} \chi_0 = 0, \quad (12)$$

Taking the inner product of the above with χ_0 , we find

$$\langle \chi_0 | \hat{A}^\dagger \hat{A} | \chi_0 \rangle = \| \hat{A} | \chi_0 \rangle \|^2 = 0 \implies \hat{A} | \chi_0 \rangle = 0. \quad (13)$$

Solving the above, we find

$$\chi_0(x) = \frac{N}{\cosh x}, \quad (14)$$

where N is a normalisation constant.

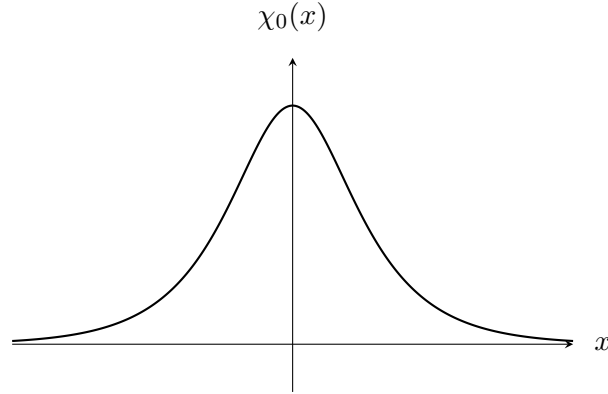


Figure 2. Ground state wavefunction $\chi_0(x) = N \operatorname{sech} x$, peaked at the origin and decaying exponentially as $x \rightarrow \pm\infty$.